

Introducing an evolutionary algorithm for the optimal training of RBF networks

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Abstract

A large collection of real - world classification and regression problems can be addressed using machine learning tools such as, for example, radial basis function networks (RBF networks). However, the techniques used for training RBF networks often exhibit various problems, such as getting trapped in local minima of the error function, or even encountering numerical issues when solving systems of linear equations in order to estimate the parameters of the RBF network. This paper presents a multi-stage evolutionary technique based on genetic algorithms for the effective training of RBF networks. In the first stage, the value ranges of the RBF network parameters are estimated using the K-Means algorithm. In the second stage, the chromosomes of the genetic algorithm are initialized within the parameter ranges determined in the first stage, followed by the execution of the genetic algorithm. Each chromosome of the genetic algorithm is considered a candidate parameter vector for the machine learning model. The centers and variances of the RBF network are estimated by the Genetic Algorithm, while the network weights are determined by solving a system of linear equations. This method was applied to a large set of classification and data-fitting problems, yielding excellent results.

Keywords: Machine learning; neural networks; genetic algorithms.

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4.1.1. Subsubsection

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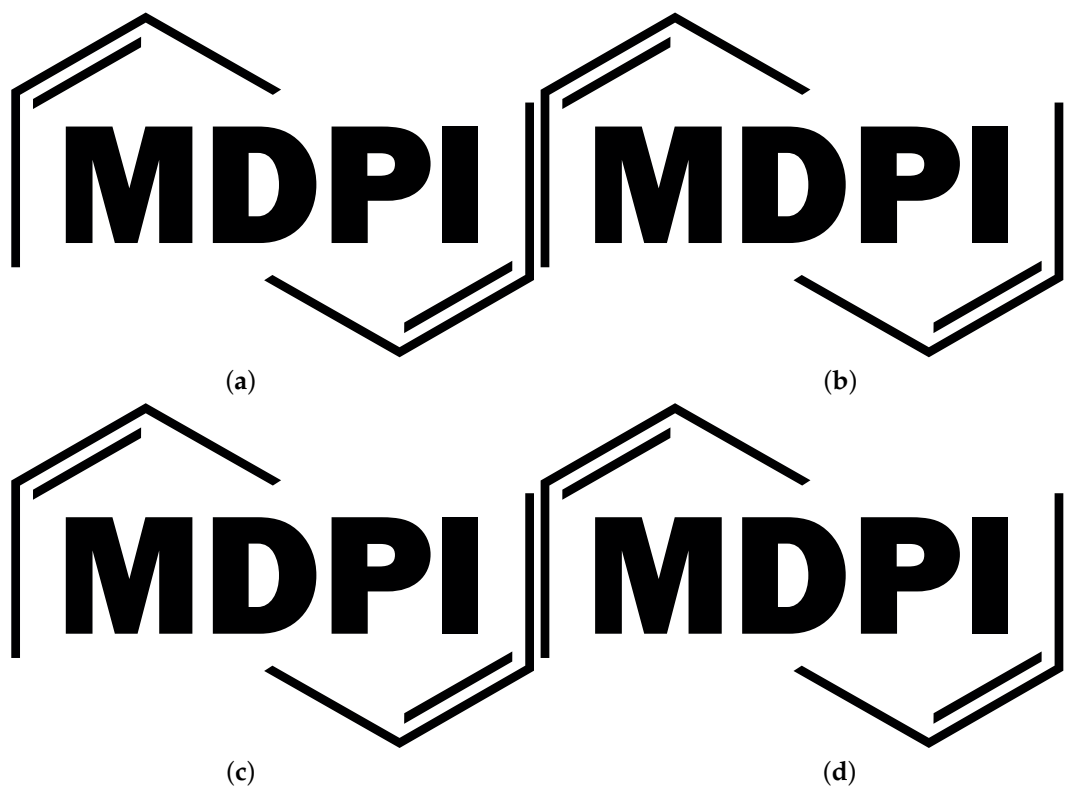


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	Data	Data	Data
	Data	Data	Data
Entry 2	Data	Data	Data
	Data	Data	Data
	Data	Data	Data

* Tables may have a footer.

Text.

Text.

80

81

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82

This is the example 1 of equation:

83

$$a = 1,$$

84

(1)

the text following an equation need not be a new paragraph. Please punctuate equations as regular text.

85

86

This is the example 2 of equation:

87

$$a = b + c + d + e + f + g + h + i + j + k + l + m + n + o + p + q + r + s + t + u + v + w + x + y + z$$

88

(2)

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Abbreviations

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LD	Linear dichroism

Appendix A

Appendix A.1

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Entry 1	Data	Data
Entry 2	Data	Data

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References

1. Author 1, T. The title of the cited article. *Journal Abbreviation* **2008**, *10*, 142–149.

2. Author 2, L. The title of the cited contribution. In *The Book Title*; Editor1, F., Editor2, A., Eds.; Publishing House: City, Country, 2007; pp. 32–58.

3. Author 1, A.; Author 2, B. *Book Title*, 3rd ed.; Publisher: Publisher Location, Country, 2008; pp. 154–196.

4. Author 1, A.B.; Author 2, C. Title of Unpublished Work. *Abbreviated Journal Name* year, phrase indicating stage of publication (submitted; accepted; in press).

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